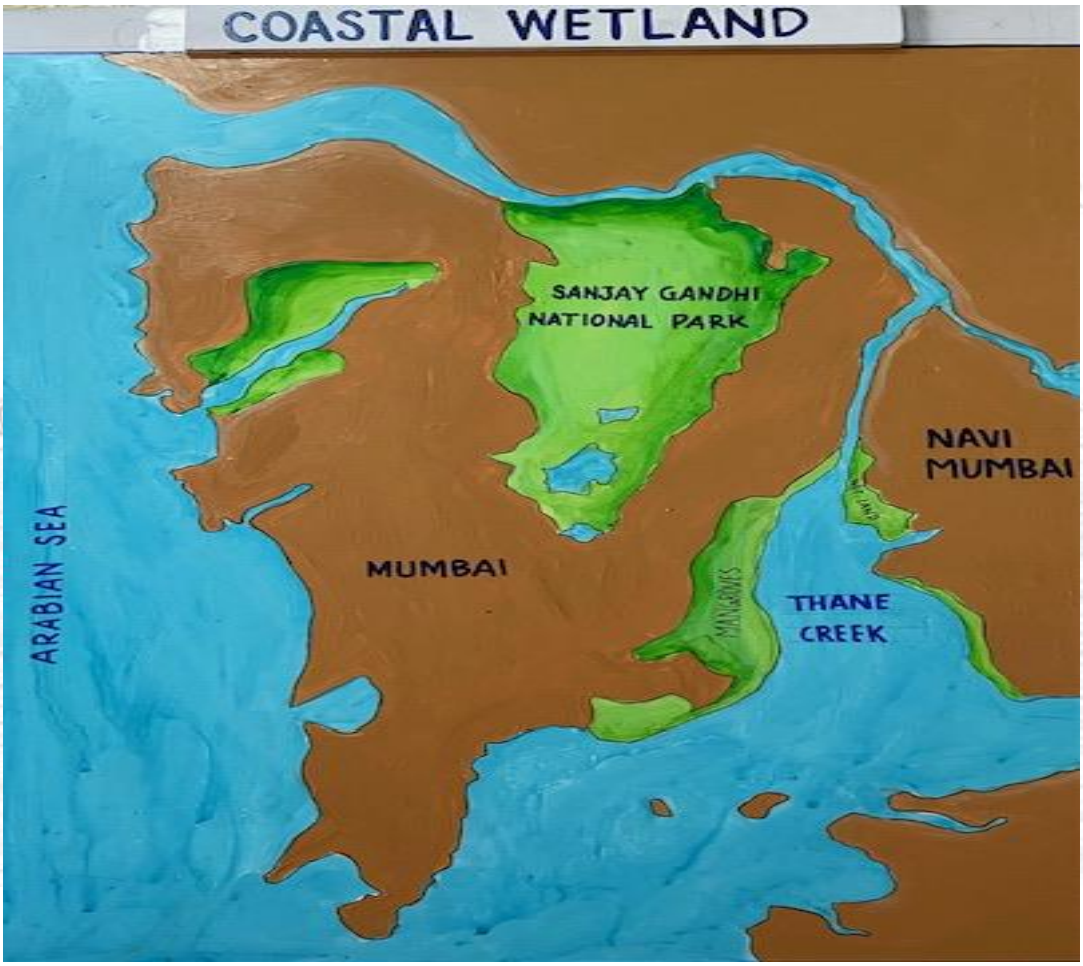
LiFE

STORY OF WETLANDS

WETLANDS UNVEILED-BLUE CARBON, SOIL CONSERVATION AND INNOVATIVE SOLUTIONS

(CBSE SCIENCE EXHIBITION 2023-24)



SCHOOL- DAV PUBLIC SCHOOL AIROLI, NAVI MUMBAI

ISHANI MAJUMDAR DEV BAJAJ

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ACKNOWLEDGEMENT

We extend our heartfelt gratitude to Dr. Hemant Karkhanis of Godrej and Boyce Mangroves for his invaluable guidance and consistent assistance in clearing our doubts throughout this research endeavor. Special thanks to Godrej Mangroves for generously allowing us access to their premises, enabling us to collect essential samples for our study.
FIE Sumeet Majumdar deserves recognition for his mentorship and unwavering support at every stage of our research.

We would like to express appreciation to the teachers at DAV Public School Airoli, lab assistants and gardener whose guidance laid a strong foundation for our academic pursuits.

We extend gratitude to the Thane Creek Flamingo Sanctuary for granting us permission to collect samples, contributing significantly to the richness of our study.

This project's success is a testament to the collaborative efforts and support received from these individuals and institutions.

Additionally, we would like to convey our thanks and special gratitude to our principal, Mrs. Suman Pradhan, for her unwavering support. We also appreciate our mentor, Jai Parkar Ma'am, for her excellent guidance and mentorship; without her, this project wouldn't have been possible. Their encouragement and leadership have been instrumental in our academic journey and the successful completion of this project.

Ishani Majumdar **Dev Bajaj**

OBJECTIVE

The primary objective of this project is to contribute to the United Nations' Sustainable Development Goals (SDGs), specifically targeting SDG 13 (Climate Action) and SDG 15 (Life on Land). Additionally, the project aims to raise awareness about the crucial role of wetlands in carbon sequestration, biodiversity preservation, and sustainable energy generation, fostering a deeper understanding of their significance for environmental conservation and global sustainability.

Synopsis:

This project delves into the critical role of wetlands, focusing on carbon sequestration, biodiversity preservation, and sustainable energy generation through the Wetland Plant Microbial Fuel Cell (PMFC). Aligned with the United Nations' Sustainable Development Goals (SDGs), our initiatives aim to address climate action, promote life on land, and contribute to affordable and clean energy solutions.

1. The project **explores the richness of wetland soils**, comparing their unique features with normal soils.
2. **Filtration experiments** demonstrate wetlands' natural cleansing capabilities, emphasizing their significance for water quality.
3. The Wetland Microbial Cell (WMC) showcases a pioneering approach to eco-friendly energy production, harnessing the power of plant-microbe interactions.
4. **Farming in wetlands** demonstrates the viability of cultivating diverse crops while **mitigating soil erosion and flood risks**. In essence, this project advocates for the protection of wetlands, positioning them as vital components in achieving a harmonious coexistence between environmental conservation and sustainable development.

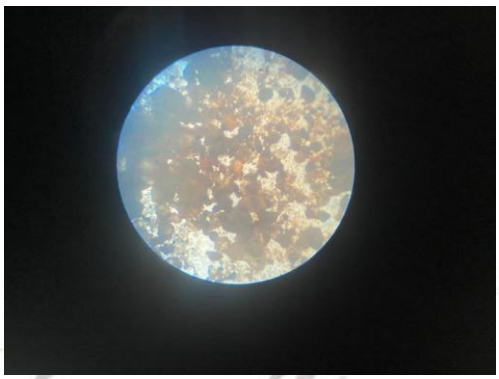
Soil Comparison and Carbon Sequestration (Blue Carbon)

Aim:

To compare wetland soils with normal soils, evaluating their unique features, impact on water dynamics, microbial activity, nutrient cycling, and, most importantly, their role in carbon sequestration, particularly storing blue carbon.



Wetland Soil under microscope



Normal soil under microscope

Steps:

- Examine wetland soils under a microscope to identify distinctive features such as increased organic matter and anaerobic conditions.
- Assess the dynamic environment created by wetland soils, focusing on water retention capabilities
- Investigate microbial activity within wetland soils, emphasizing redox reactions and their influence on nutrient cycling.
- Explore the capacity of wetland soils to sequester carbon, specifically blue carbon, through a comprehensive study of coastal and marine ecosystems.
- Analyse and contrast the characteristics and functions of wetland soils with those of normal soils.

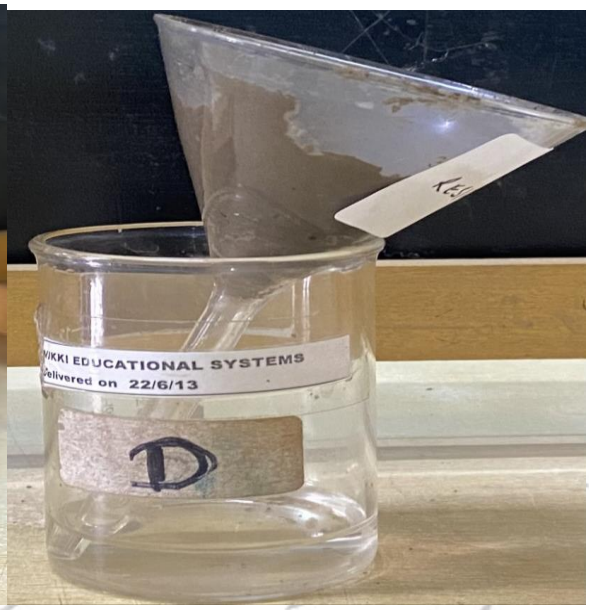
Inference:

Wetland soils, with their unique features, play a pivotal role in water dynamics, microbial activity, and nutrient cycling. They emerge as crucial carbon sinks, notably storing blue carbon, contributing significantly to climate change mitigation globally.

Kidneys of Landscapes (Filtration)

Aim: To demonstrate the filtration of substances using the wetland soil collected from mangroves and evaluating their role in enhancing the water quality for coastal waters.





BEFORE FILTRATION **AFTER FILTRATION**

Steps:

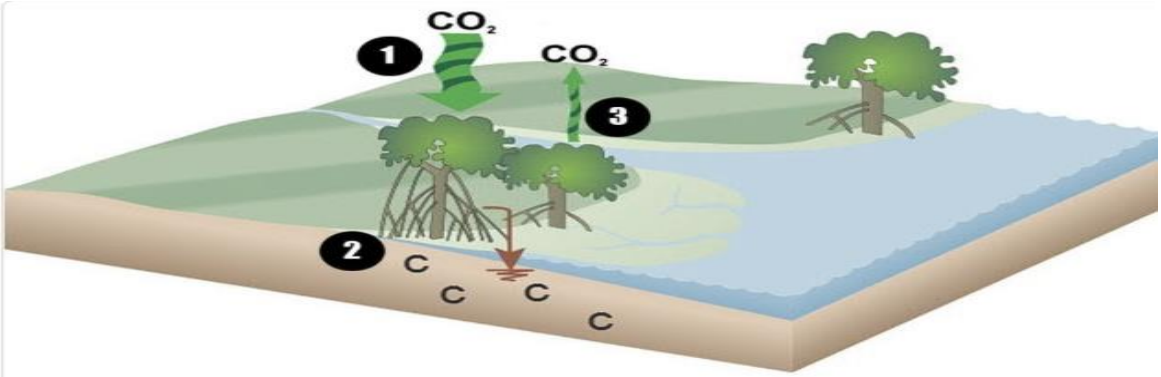
- Use the wetland soil samples in a funnel and pass impure water through it.
- Wait for the filtration process.

Inference:

- The filtrate has better clarity and quality.The soil (residue) absorbs the excess nutrients and harmful microbes, preventing many coastal water safety issues.
- Wetlands soil has a unique property of efficient filtration of harmful contents such as nitrogen, phosphorous, heavy metals and other microbes to prevent coastal water issues such as algal blooms. The filtrate i.e clear water after treatment can be used for farming, industries and other purposes. Solving the freshwater crisis. Thus, wetlands act as kidneys of landscape.

Blue Carbon Sequestration:

Aim: To show how water captures and stores carbon dioxide in the ecosystem using NaHCO3, COOCH3 and NAOH.



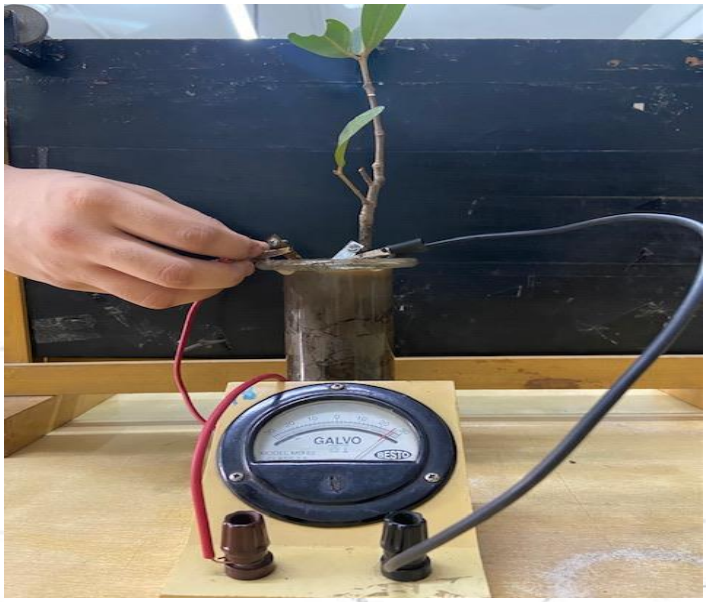
This diagram illustrates the mechanisms by which carbon moves into and out of coastal wetlands: **(1)** Carbon dioxide in the atmosphere is taken in by trees and plants during the process of photosynthesis. This is called sequestration. **(2)** Dead leaves, branches, and roots containing carbon are buried in the soil, which is frequently, if not always, covered with tidal waters. This oxygen-poor environment causes very slow breakdown of the plant materials, resulting in significant carbon storage. **(3)** A small amount of carbon is lost back to the atmosphere through respiration, while the rest is stored in the leaves, branches, and roots of the plants. *This diagram is adapted from a figure in Sutton-Grier et al. 2014 Marine Policy*

Wetland soils excel in carbon removal, storing 81 to 216 metric tons of carbon per acre, contingent on their type and location. This underscores their unparalleled capacity as carbon sinks and emphasizes their critical role in mitigating climate change. Wetlands, as custodians of blue carbon, highlight the intricate connections between local and global environmental processes, underscoring the urgency of their protection for Earth's ecosystems and climate resilience.

Wetland Microbial Cell (WMC):

Aim:

To harness sustainable electrical energy through the Wetland Plant Microbial Fuel Cell (WMC) by leveraging plant-soil-microbe interactions.



Steps:

- Initiation: As plant roots release organic compounds into wetland soil, indigenous microbes participate in redox reactions.
- Electron Release: Microbes release electrons during redox reactions, creating an electrical potential difference.
- Capture Process: The WMC captures electrons through an anode immersed in the wetland soil, establishing a potential difference with a cathode.
- Energy Generation: The potential difference yields electrical energy as a result of microbial-electrochemical synergy.

Inference:

The Wetland PMFC demonstrates a sustainable and eco-friendly energy solution, capitalizing on natural resources within wetlands. This technology offers a cleaner alternative to conventional fuel sources, producing no greenhouse gases or pollutants. Scaling up WMCs for large-scale deployment holds promise for decentralized power generation, especially in remote or environmentally sensitive areas.

Wetland Farming:

AIM:

To show tidal wetlands containing clayey soil can be utilised for paddy farming.



Steps:

- Soak the paddy in water for 2 days and keep it wrapped in a wet cloth for 2 more days for germination.
- Sow the germinated saplings in wetland clayey soil from mangroves for growing.

Inference:

Tidal wetlands have less intrusion of saline water hence helps in growing of crops such as paddy, sugarcane and jute.

Wetland farming, also known as wetland agriculture or paddy farming, involves the cultivation of crops in areas with waterlogged or saturated soil conditions. The most common form of wetland farming is rice cultivation, where rice paddies are flooded to create the ideal conditions for rice plants.

Studies show that - Wetland farming, when managed sustainably, can contribute to food security, livelihoods, and environmental conservation. However, it is important to balance agricultural activities with the preservation of wetland ecosystems to ensure long-term sustainability and resilience in the face of environmental changes.

SOIL CONSERVATION:

Aim:

To show binding of soil by wetland plants resulting in conservation of soil erosion.



Steps:

- Pull the red mangrove plant from the clayey soil.
- Observe the dense roots of the plant and the soil bound to it.

Inference:

- The dense root systems of wetland plants, such as marsh grasses and other aquatic vegetation, contribute to several aspects of soil conservation.
- The extensive and dense root systems of wetland plants help bind the soil particles together, providing stability to the soil. This prevents soil erosion caused by water flow and wave action.
- Conserving wetlands and their dense root systems is essential for maintaining these ecosystem services. Human activities that disrupt wetland habitats, such as draining for agriculture or urban development, can have significant negative impacts on soil conservation and the overall health of wetland ecosystems. Therefore, it is crucial to implement conservation measures and sustainable land management practices to protect and restore wetlands.

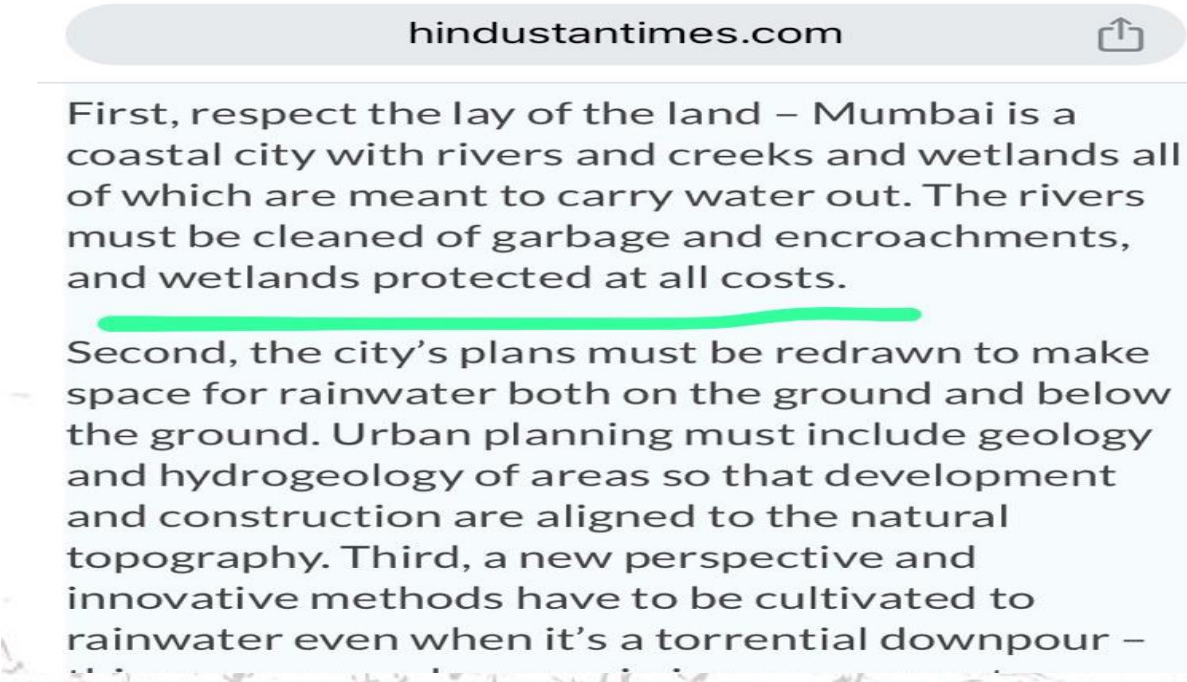
FIELD TRIP


Field Visit: Thane Creek
THANE CREEK
FLAMINGO
SANCTUARY
Airoli
Implemented by
giz
Thane Creek

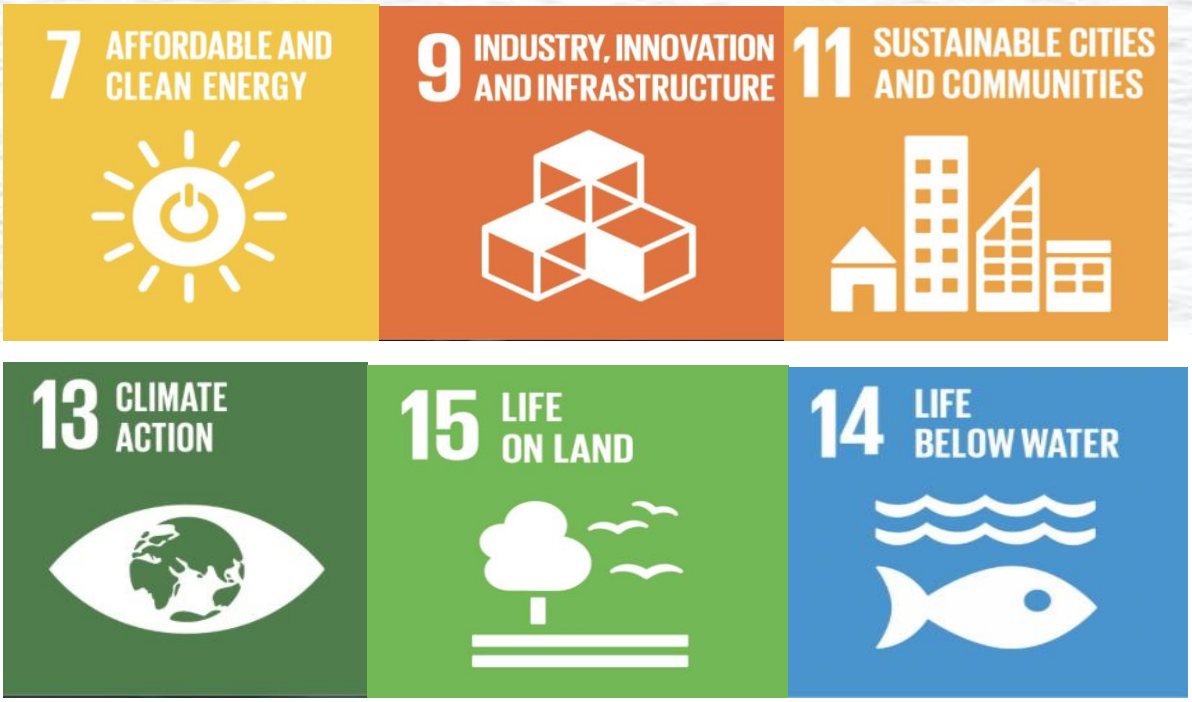

WATCH
TOWER
Orange Mangrove

NOTE:

Reducing the impact of flooding in cities due to protection of URBAN WETLANDS



SDGs Achieved through this Project:



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